Experiment 3

3D CAD Modelling of a Forked Bracket with Cylindrical Projection

Aim

To design and model a forked bracket component in 3D CAD software using dimensioned 2D sketches, extrusion, and solid-modelling operations.

Requirements

• Computer with 3D CAD software (such as AutoCAD or an equivalent solid-modelling application)
• Mouse and keyboard
• Supplied dimensioned reference sketches and model images

Procedure

1. Draw the profiles and enter the dimensions. Begin by creating the required 2D sketches on the appropriate planes. Use the dimensions visible in the reference images: rectangular face 40 mm wide × 32 mm high with a central Ø18 mm circular feature; fork/end profile with R16 mm rounded end and Ø16 mm hole; 36 mm dimension shown along the body; and the cylindrical projection with Ø18 mm diameter and 40 mm length/height as indicated in the side view. Keep the profiles aligned to the sketch axes and position the holes centrally in their respective rounded sections.

2. Prepare the forked body sketch. Draw the U-shaped/forked outline with two parallel arms and the connecting block. Add the rounded end and hole profile, then check the spacing and alignment against the supplied sketches.

Dimensions

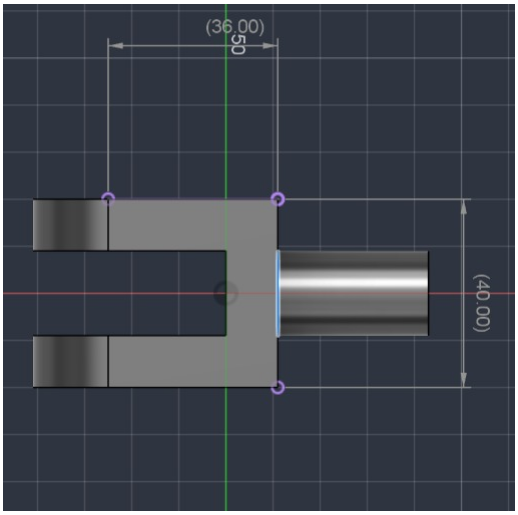


Figure 1. Side/profile dimensions showing the 36 mm and 40 mm measurements.

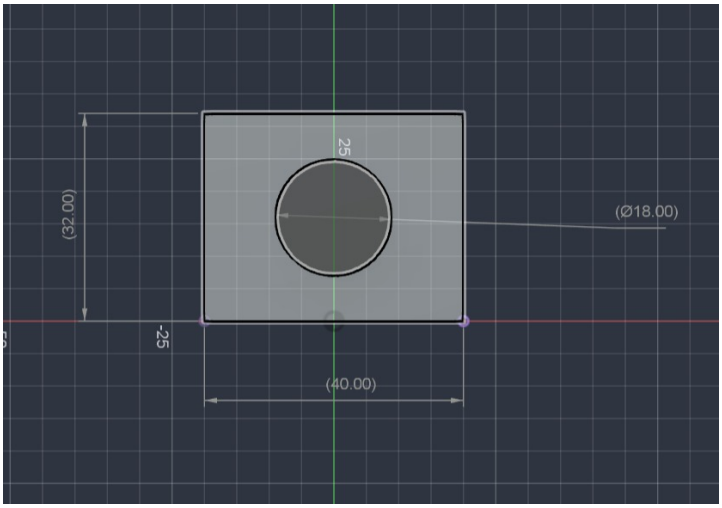


Figure 2. Face dimensions showing the Ø18 mm circular feature and 40 × 32 mm outline.

Dimensions and Procedure (continued)

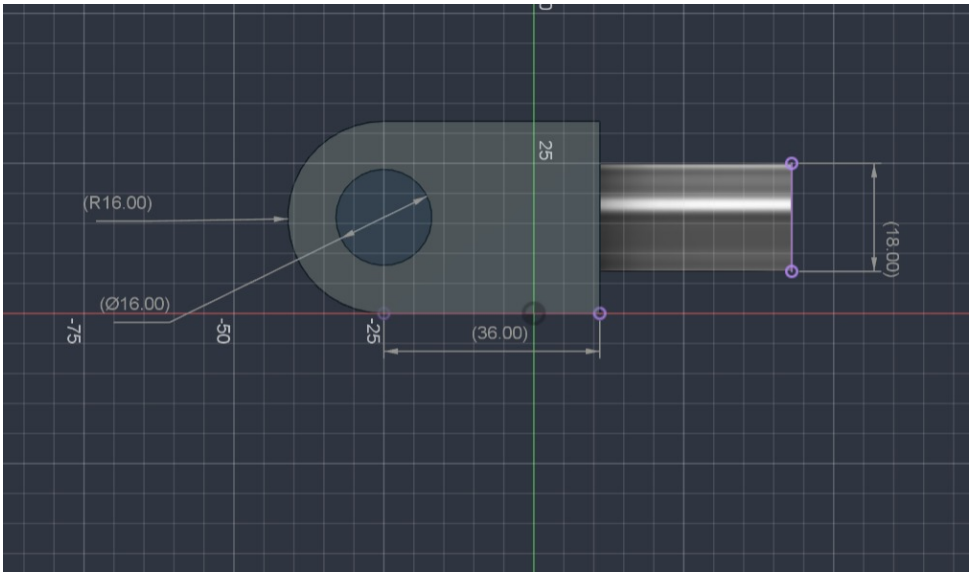


Figure 3. Fork-end dimensions showing R16 mm, Ø16 mm, and the 36 mm measurement.

3. Extrude the profiles. Select the closed sketch regions and apply the required extrusion depths to form the solid bracket. Extrude the forked arms and connecting body to create the main solid shape. Then create the cylindrical projection from the circular profile, using the Ø18 mm diameter and the length shown in the reference. The extruded result is shown in Figure 4.

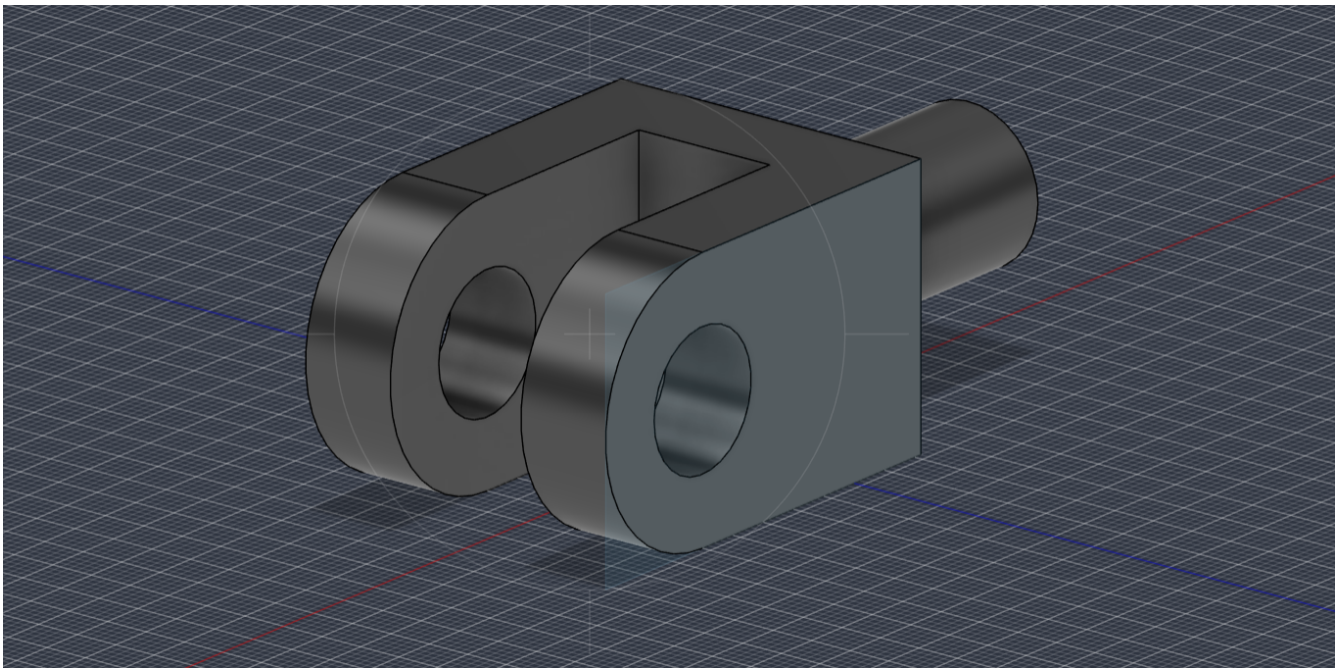


Figure 4. Extruded 3D model showing the forked bracket and cylindrical projection.

Procedure (continued)

4. Inspect and refine the model. Review the isometric and orthographic views. Confirm that the fork gap, rounded ends, holes, connecting body, and cylindrical projection are formed correctly. Make any required adjustments before preparing the final drawing sheet.

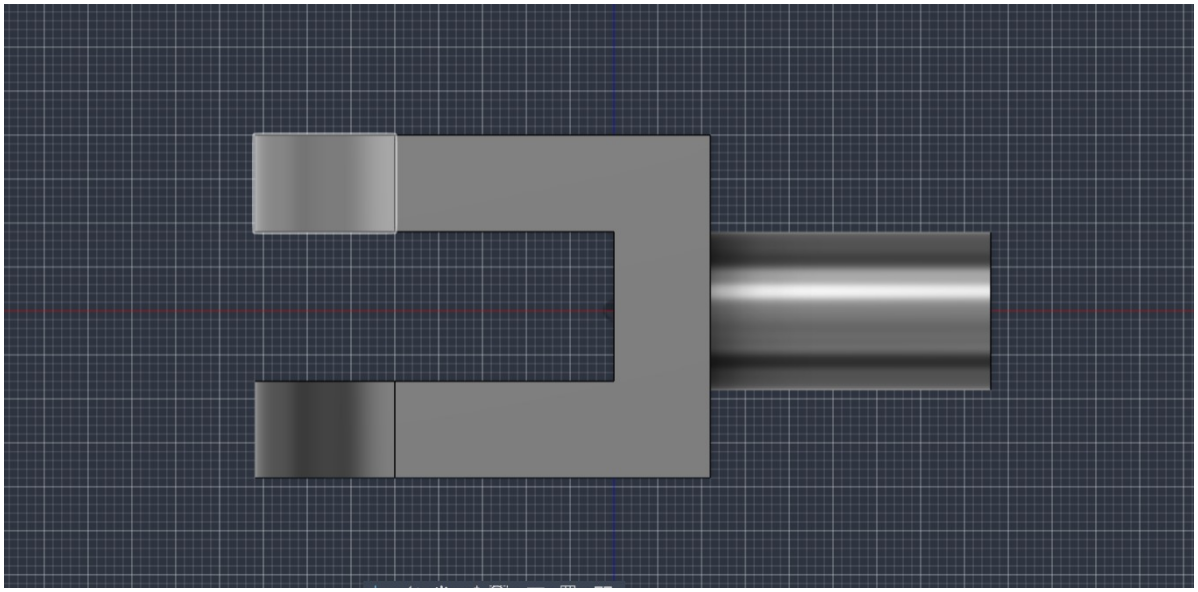


Figure 5. Orthographic/top view of the completed solid model.

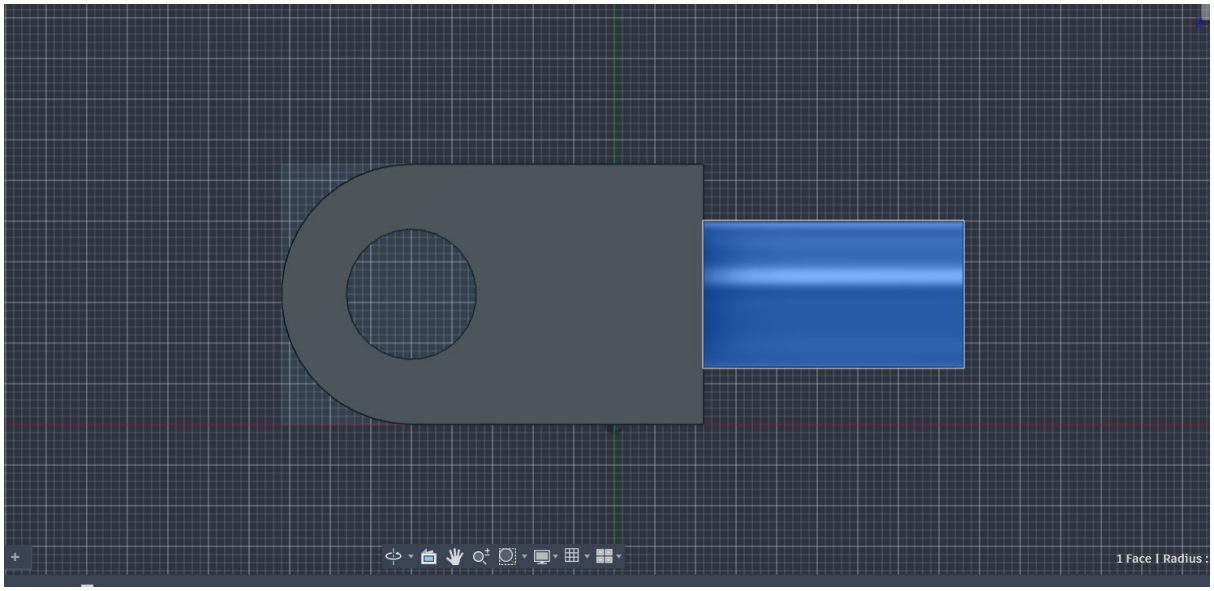


Figure 6. Additional view used to verify the forked shape and cylindrical projection.

Output

The final output is the last supplied picture: the completed engineering drawing sheet. It presents the finished component through orthographic and isometric views, showing the forked arms, circular holes, main body, and cylindrical projection.

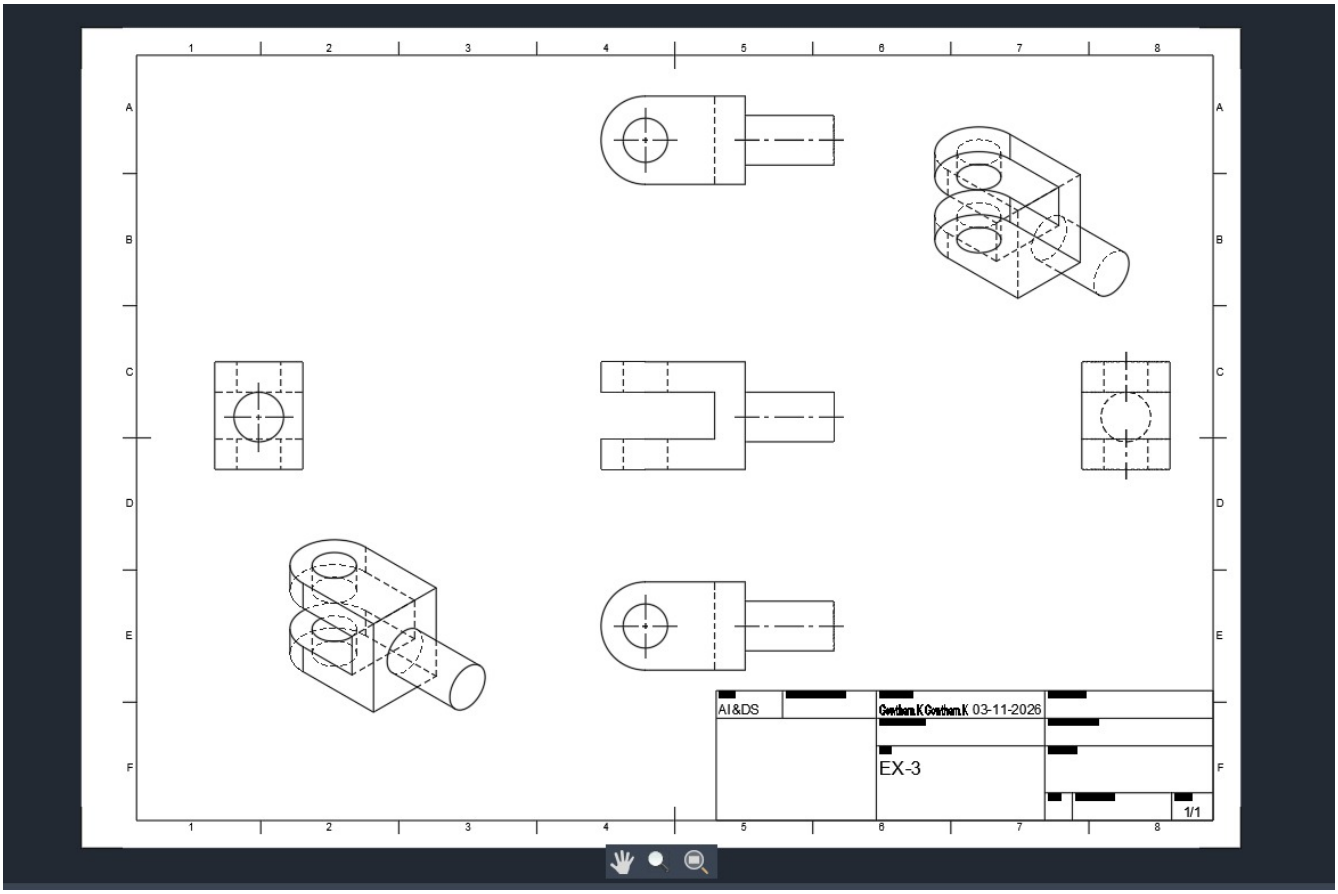


Figure 7. Final drawing sheet (last reference image) showing the completed Experiment 3 model in multiple views.

Result

The forked bracket with a cylindrical projection was successfully modelled using dimensioned sketches and extrusion operations. The final drawing sheet displays the completed geometry in multiple views for clear visualization and verification.